Signed Simpson Calculus for Reflected-Clock Iterated Brownian Motion

Project Stardust
Private working exposition

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Abstract

We summarize verified stochastic constructions for the actual process $Z_t = W(|B_t|)$ on the existing independent scenery–clock probability space. Its uniform-grid quartic variation converges in probability to $3T$, whereas quadratic variation diverges. Separately convergent raw first and second Taylor channels already fail for the quadratic polynomial. Signed spatial Simpson sums nevertheless converge for every global C^3 one-form, with uniform control over grid prefixes, canonical bounded random horizons, and deterministic C^1 time weights. An explicit clipped cosine construction converges in observable L^2 and in the original intrinsic graph dual, while its limit stays outside the strong joint derivative domain. These are specified symmetric formulas, not completion of adapted physical-time Itô–Tanaka calculus.

Verification contract. The mathematical baseline is the fully audited private commit

`072793e54f7a08ba39afc4919ea987d11e5c69e2`.

Results below summarize checkpoints 1383–1394. Every cited name has prefix

`OperatorDerivativeRewrite.Foundation.`

The original manuscripts are preserved unchanged. This exposition asserts no external novelty, publication readiness, or completion of the IBM campaign.

1 Process, prescription, and topology

Let W be the constructed continuous Brownian scenery on nonnegative spatial time and B the independent constructed Brownian clock. The process studied here is precisely

$$Z_t = W(|B_t|), \qquad t \geq 0, \tag{1}$$

under the original product probability law. This fixes the reflected-clock convention; no equality with every other convention for iterated Brownian motion is asserted. Its paths are continuous for every sample in the chosen realization, and evaluation is jointly measurable. We retain Z_0 in finite identities and targets rather than simplify arbitrary L^2 representatives.

Fix $T > 0$ and $N = n + 1$. Write $t_i = iT/N$, $Z_i = Z_{t_i}$ and $\Delta_i Z = Z_{i+1} - Z_i$. Unless stated otherwise, stochastic limits are in probability under this same original law, along all $n \to \infty$. Samplewise identities are distinguished from probability limits.

2 Variation and a concrete obstruction

Theorem 2.1 (Actual uniform-grid variation)**.** *For each fixed positive T,*

$$Q_{4,N} := \sum_{i<N} (\Delta_i Z)^4 \xrightarrow{\mathbf{P}} 3T, \qquad \mathbf{P}\left\{ \sum_{i<N} (\Delta_i Z)^2 \le K \right\} \longrightarrow 0 \quad (K \in \mathbb{R}).$$

The maximum absolute grid increment tends to zero on every sample.

Lean reference: `IBMUniformQuarticVariation`
`quartic_tendstoInMeasure`

Lean reference: `IBMUniformFifthRemainder`
`mesh_tendsto`

Lean reference: `IBMActualQuadraticDivergence`
`quadratic_lower_tail`

Verified mechanisms. The quartic theorem uses the actual conditional Gaussian contractions, overlap estimates and reflected-clock crossing penalty. It is not inferred from its expectation alone. The deterministic inequality

$$Q_{4,N} \le \left(\max_{i<N} |\Delta_i Z| \right)^2 Q_{2,N}$$

then forces divergence of $Q_{2,N}$ in probability. The formal proof first shows $(1 + Q_{2,N})^{-1} \xrightarrow{\mathbf{P}} 0$ and controls each lower-tail event by a strictly positive reciprocal threshold. No almost-sure full-sequence divergence or divergence rate is claimed. □

For $f(z) = z^2$, the two raw ranks satisfy, on every grid and sample,

$$L_N + Q_{2,N} = Z_T^2 - Z_0^2, \qquad L_N = \sum_{i<N} 2 Z_i \Delta_i Z. \tag{2}$$

Neither L_N nor $Q_{2,N}$ has any finite real-valued random probability limit. Thus convergence of a combined channel expression does not license separate unrenormalized integrals.

Lean reference: `IBMActualQuadraticDivergence`
`first_identity`

Lean reference: `IBMActualQuadraticDivergence`
`first_no_finite_limit`

3 Combined Taylor formula and signed Simpson sums

Theorem 3.1 (Combined lower Taylor channels)**.** *For every global C^4 real function f,*

$$\sum_{i<N} \left[f'(Z_i)\Delta_i Z + \frac{1}{2} f''(Z_i)(\Delta_i Z)^2 + \frac{1}{6} f'''(Z_i)(\Delta_i Z)^3 \right] \xrightarrow{\mathbf{P}} f(Z_T) - f(Z_0) - \frac{1}{8} \int_0^T f^{(4)}(Z_t)\, dt.$$

No fifth derivative or globally bounded derivative is assumed.

Lean reference: `IBMActualTaylorChain.channels_tendstoInMeasure`

The coefficient $1/8$ is $3/24$: the actual continuously weighted quartic limit supplies $3\int f^{(4)}(Z_t)dt$, while the fourth Taylor coefficient is $1/24$. Compact Peano control proves decay of the total absolute Taylor error on the entire C^4 domain. The channels in Theorem 3.1 remain together.

Define a signed increment using only its integrand and spatial endpoints:

$$S_g(x,y) = \frac{y-x}{6}\left[g(x) + 4g\left(\frac{x+y}{2}\right) + g(y)\right]. \tag{3}$$

The middle argument is a *spatial* midpoint, not Z at a middle physical time. The sign of $y-x$ is retained.

Theorem 3.2 (Actual symmetric spatial integration)**.** *For every global C^3 real function g,*

$$\sum_{i<N} S_g(Z_i, Z_{i+1}) \xrightarrow{\mathbf{P}} J_g(T), \qquad J_g(t) = \int_{Z_0}^{Z_t} g(z)\,dz. \tag{4}$$

Each target is a genuine finite oriented Lebesgue integral. For continuous g, J_g is already a jointly measurable process with continuous paths and a measurable realization in continuous-path space.

Lean reference: `IBMActualSimpson.sums_tendstoInMeasure`

Lean reference: `IBMUniformSimpsonProcess.integralPath_measurable`

Verified mechanism. For $f \in C^4$, the Simpson error relative to $f(y) - f(x)$ is uniformly $o(|y-x|^4)$ on compact spatial intervals. This follows from compact Peano control for f and for the third-order Taylor remainder of f'. Every sample has a compact range and vanishing grid mesh. Total error is therefore eventually bounded by $\varepsilon Q_{4,N}$ for each $\varepsilon > 0$. Quartic tightness and the subsequence criterion give probability convergence. For general $g \in C^3$, the independently defined ordinary primitive $f(x) = \int_0^x g$ is C^4 by FTC and has derivative g. □

This is convergence of an actual approximation, not an endpoint definition of the finite sums. For example,

$$S_{5z^4}(x,y) - (y^5 - x^5) = \frac{(y-x)^5}{24},$$

which is generally nonzero. In contrast, $S_{4z^3}(x,y) = y^4 - x^4$ exactly. For $f(z) = z^4$, Theorem 3.1 has correction $3T$, whereas the Simpson rule for f' telescopes exactly. These are distinct prescriptions, consistent with the cancellation in (2).

4 Uniform prefixes and bounded random horizons

Let $P_{g,N}(m) = \sum_{i<m} S_g(Z_i, Z_{i+1})$ on the same N-cell grid.

Theorem 4.1 (Uniform grid-prefix realization)**.** *For every global C^3 one-form,*

$$\max_{0\le m\le N} |P_{g,N}(m) - J_g(mT/N)| \xrightarrow{\mathbf{P}} 0.$$

Lean reference: `IBMUniformSimpsonProcess`
`maximum_tendstoInMeasure`

One compact modulus controls every prefix; its quartic prefix mass is at most the full nonnegative quartic mass. No union bound growing with N is used. This is uniformity at *grid prefixes*. No continuous interpolation of the finite sums, or uniform real-time approximation theorem, is supplied.

Theorem 4.2 (Canonical bounded random endpoint)**.** *Let τ be any measurable nonnegative time with the pointwise bound $\tau \le T$. Put*

$$k_N(\tau) = \left\lfloor \frac{N\tau}{T} \right\rfloor, \qquad \tau_N = \frac{T}{N} k_N(\tau).$$

Then $k_N(\tau) \le N$, $0 \le \tau - \tau_N < T/N$, and $P_{g,N}(k_N(\tau)) \xrightarrow{\mathbf{P}} J_g(\tau)$ for every global C^3 one-form. The approximants and target are measurable.

Lean reference: `IBMSimpsonRandomHorizon`
`stopped_sums_tendstoInMeasure`

Uniform prefix control and every-sample continuity of J_g prove the result. The time need not be a stopping time; an example chooses T or $T/2$ according to the sign of Z_T. This is not optional sampling of a martingale.

5 Deterministic time weights and spacetime chains

Theorem 5.1 (Actual time-weighted formula)**.** *For global $h \in C^1(\mathbb{R})$ and $g \in C^3(\mathbb{R})$,*

$$\sum_{i<N} h(t_i) S_g(Z_i, Z_{i+1}) \xrightarrow{\mathbf{P}} h(T) J_g(T) - \int_0^T h'(t) J_g(t)\, dt.$$

The time correction is finite on every sample and measurable. No global growth bound or expectation moment is assumed.

Lean reference: `IBMTimeWeightedSimpson`
`weighted_tendstoInMeasure`

Verified mechanism. Finite Abel summation bounds the error in replacing Simpson increments by actual increments of J_g by $M_N(B + KT)$, where M_N is the prefix maximum from Theorem 4.1, $|h| \le B$ on $[0, T]$, and h has compact Lipschitz bound K. These constants follow from C^1 regularity, not global assumptions. Finite product telescoping identifies the remaining path sum. FTC and uniform continuity of each continuous sample path give $\sum J_g(t_{i+1})\Delta_i h \to \int h' J_g$ samplewise. The time integral is proved directly through cell integrals, rather than assumed as a formal Riemann–Stieltjes limit. □

In particular, for $f \in C^4$,

$$\sum_{i<N} h(t_i) S_{f'}(Z_i, Z_{i+1}) \xrightarrow{\mathbf{P}} h(T) f(Z_T) - h(0) f(Z_0) - \int_0^T h'(t) f(Z_t)\, dt.$$

Lean reference: `IBMTimeWeightedSimpson`
`derivative_weighted_tendstoInMeasure`

This is a separable spacetime chain rule. Examples tZ_t, tZ_t^2, and $t^2 \sin(Z_t)$ are independently tested in Lean. General two-variable $F(t, z)$ and random time coefficients remain separate obligations.

6 A precise graph-dual connection and strong-domain limit

Let D be the existing closed, densely defined intrinsic joint derivative on the original observable L^2, and let $\delta = D^*$ denote its original Hilbert adjoint. Its full adjoint-domain graph tests support the pre-existing bounded graph-dual map $\mathfrak{D}$. Write emb for the existing embedding of strong two-channel fields into that test dual.

Take the actual cosine Simpson sum $A_N = \sum S_{\cos}(Z_i, Z_{i+1})$ and define

$$C_N = \max(-1, \min(1, A_N)).$$

Clipping is an explicit modification: already $S_{\cos}(0, 2\pi) = -2\pi/3$. It is 1-Lipschitz and fixes the sine target, so it preserves convergence in probability. The bounded actual functions C_N are IBM observables.

Theorem 6.1 (Bounded realization in the original graph dual)**.** *For every $T > 0$, $C_N \to \sin(Z_T)$ strongly in observable L^2 and*

$$\mathfrak{D}C_N \longrightarrow \mathfrak{D}\sin(Z_T) \quad \textit{in the original graph-dual norm.}$$

For every $v \in \mathrm{Dom}(\delta)$, the actual integrable pairings obey

$$\mathbf{E}[C_N\, \delta v] \longrightarrow (\mathfrak{D}\sin(Z_T))(\mathrm{testOf}\, v).$$

Lean reference: `IBMSimpsonGraphDual`
`valueClass_tendsto`

Lean reference: `IBMSimpsonGraphDual`
`actual_pairing_tendsto`

Bounded dominated convergence on a.e. subsubsequences supplies strong L^2 convergence; continuity of the original graph-dual map and its evaluations then supplies the stated derivative and pairing limits. No L^2 convergence of the *unclipped* sums is asserted.

The previously verified sine observation is nevertheless outside $\mathrm{Dom}(D)$ for every positive T; its graph-dual derivative is nonzero and is not $\mathrm{emb}(V)$ for any strong joint field V. Any strong-domain sequence tracking C_N in L^2 has embedded gradients converging in the graph dual, but its strong gradient norms tend to infinity. The tracking premise is explicit; the finite C_N themselves are not asserted to lie in the strong domain.

Lean reference: `IBMSimpsonGraphDual`
`limit_obstruction`

Lean reference: `IBMSimpsonGraphDual`
`tracked_graph_escape`

7 What is established and what remains

The results construct specified signed quadratures, genuine process-valued targets, grid-prefix and bounded random-horizon convergence, a deterministic time-weighted chain formula, and an exact bounded graph-dual realization. They do not identify the spatial one-form with a Skorohod integrand or the Simpson integral with a causal divergence.

Existing predictable integration on the IBM observation filtration uses a separately constructed terminal-observable martingale. Sharing a filtration does not identify that integrator with physical

Z. The earlier change-of- integrator draft concerns that verified martingale calculus, and remains valid on its own original domains.

Open repository obligations include general $F(t, z)$, controlled random time-dependent coefficients, continuous finite-sum interpolation and its real-time uniform topology, arbitrary partitions, required causal or adjoint identifications, and nonsmooth physical-time Itô–Tanaka formulas. Strong domain exclusions cannot be repaired by silently assuming membership. The verified stopping contract for the full IBM campaign has not been met.

All statements refer to pinned Lean 4.34.1 and the existing Mathlib revision. The baseline complete transitive audit checks all safe Foundation declarations and admits only `propext`, `Classical.choice`, and `Quot.sound`. This source/PDF pair is a private working synthesis of those results, not an original manuscript from the research archive.